CLOCKED BUILD CHALLENGE · 2026

AuditTrail AI

Lease compliance with receipts.

FastAPI · Supabase · Clocked
Five endpoints · three groups
github.com/[REPO] · [DEMO_URL]

A commercial lease audit in five seconds, with an evidence quote for every finding. Paste a lease and the zoning source it must satisfy. AuditTrail masks every tenant identifier, reads the operative terms verbatim, tests each claim against the source, computes the numeric rules itself, and returns a defensible first pass that a firm can send to a client.

The problem

A first-pass lease review takes three to six hours, mostly locating clauses and cross-checking a zoning extract. Generic LLMs make it worse: they summarise without citations and cannot be handed a tenant's name, SSN and bank details.

The gap is not "an AI that reads contracts". It is an audit a lawyer can defend line by line, on documents that never leave the firm.

Why it holds up

- **Evidence, not prose.** Every claim returns supported, contradicted or not stated, with a quote and character offsets.
- **Models read, code counts.** Parking ratios, area caps, hours and consent standards are computed and shown as formulas.
- **Privacy by construction.** Stable tokens replace identifiers; the tenant and landlord stay distinguishable; the re-identification key is never stored.
- **Fails closed.** Guard screens for prompt injection before any further model call.

The five Clocked calls, and what each one changes

STEP	ENDPOINT	GROUP	OUTPUT → DOWNSTREAM EFFECT
Protect	`POST /clocked/pii`	Protect / Guard / Inspect	Entity spans become stable tokens; only masked text reaches the later steps.
Guard	`POST /clocked/injection`	Protect / Guard / Inspect	A dangerous verdict aborts the audit before extract, verify and decide.
Extract	`POST /clocked/extract`	Decide / Extract / Fill	Twelve verbatim fields become the claim list and the numeric facts.
Verify	`POST /clocked/grounding`	Verify / Compare / Rerank	Per-claim verdicts and quotes drive the risk policy and the report.
Decide	`POST /clocked/decide`	Decide / Extract / Fill	Calibrated risk, actions and review flag set the banner and checklist.

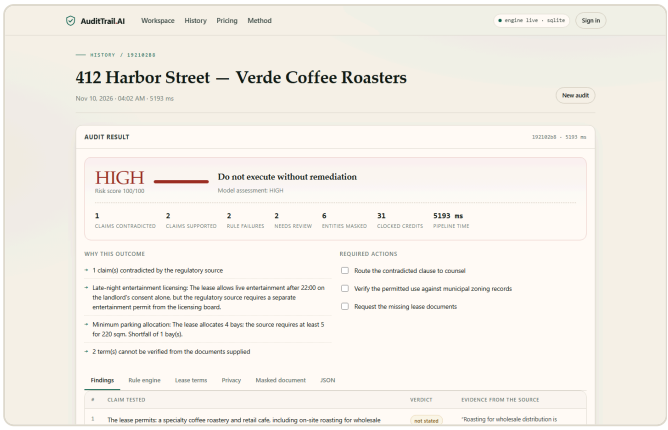
5	3	40	5–7s	24–32
CLOCKED ENDPOINTS	ENDPOINT GROUPS	TESTS PASSING	PER AUDIT	CREDIT COST

Why the arithmetic is separate. Asked whether four bays satisfy a two-per-100 sqm rule on 220 sqm, the model answered “not stated”. The rule engine computed ceil(220 ÷ 100 × 2) = 5 and reported a one-bay shortfall. Models read; code counts.

Evidence from the bundled audit

DETERMINISTIC CHECK	RESULT	COMPUTATION
Minimum parking allocation	Fail	`ceil(220 ÷ 100 × 2) = 5 vs 4`
Late-night entertainment permit	Fail	`consent alone ≠ permit`
Roasting process area cap	Review	`process area not stated`
Service bay delivery window	Pass	`05:00–11:00 ⊆ 05:00–11:00`

> "Live entertainment after 22:00 requires a supplementary entertainment permit from the District Licensing Board; the landlord's consent alone is not sufficient."
>
> Clocked Verify, evidence quote with source offsets



The result screen: risk banner, rationale, claim table and exports.

Built as a product, not a demo

ACCOUNTS & ISOLATION

Supabase Auth with row-level security; the backend verifies the token and ignores any client-supplied user id.

PLANS & QUOTAS

Free, Solo, Firm and Enterprise; server-side monthly allowance; 402 with usage for upgrades; Stripe Checkout when configured.

MACHINE ACCESS

Paid plans mint API keys, stored only as SHA-256 digests, accepted as X-API-Key on the same routes.

DELIVERABLE OUTPUT

Print-ready HTML/PDF, Markdown and JSON; white-label on Firm; every export carries the evidence and the limitations.

OPERABLE

One command to run, Docker and Render config, health endpoint, sanitised errors, SQLite fallback when Supabase is absent.

MEASURED USAGE

Each report records Clocked credits and model milliseconds, computed from the API's own timing field.

Pricing and unit economics

REVIEW

$0

- 3 audits/mo
- Full evidence report

SOLO

$49

- 100 audits/mo
- Export + API keys

FIRM

$199

- 500 audits, 5 seats
- White-label reports

ENTERPRISE

Custom

- On-prem / VPC
- SSO, rule packs

Overage $2.00 per audit. Measured cost to serve: **24–32 Clocked credits, $0.007–$0.010** (1 credit = 25 ms model time; $0.30 per 1,000 credits on Build). That is a **~98% gross margin** at Solo list price. Break-even is roughly two Solo customers.

What the judges can verify in a minute

- python run.py → header shows **engine live**.
- **Load sample** → **Run audit** → five named steps with real timings.
- Verdict **HIGH · 100**; rationale names the parking shortfall and the licensing conflict.
- Rule tab shows ceil(220 ÷ 100 × 2) = 5 required vs 4 allocated.
- Findings tab shows the contradicted claim with its quote.
- Privacy tab shows six tokens and no raw identifiers.
- Printable report renders the same audit as a client document.
- GET /api/v1/health reports engine, storage and the five steps.

Not implemented: Compare, Fill, Rerank, Inspect. The rule engine covers the numeric patterns it knows; other jurisdictions need a rule pack. Supabase needs a one-time schema run; until then the product uses SQLite and says so.